

Sturm - Liouville (S.L.) Equation

$$\left(r(x) y' \right)' + [q(x) + \lambda p(x)] y = 0$$

$r(x)$, $p(x)$, $q(x)$ are functions of x , λ is a constant.

• Legendre's Equation is a S.L. Equation

$$(1-x^2) y'' - 2xy' + n(n+1)y = 0$$

$$\Rightarrow \left[(1-x^2) y' \right]' + n(n+1)y = 0$$

$$\bullet \quad r(x) = 1-x^2, \quad q(x) = 0, \quad p(x) = 1, \quad \lambda = n(n+1)$$

Boundary value problem

$$(r(x)y')' + (q(x) + \lambda p(x))y = 0 \quad ; \quad a \leq x \leq b$$

$$\left. \begin{aligned} k_1 y(a) + k_2 y'(a) &= 0 \\ l_1 y(b) + l_2 y'(b) &= 0 \end{aligned} \right\} \begin{aligned} &k_1 \text{ and } k_2 \text{ are constant,} \\ &\text{not both zero.} \\ &l_1 \text{ and } l_2 \text{ are constant,} \\ &\text{not both zero.} \end{aligned}$$

$r(x)$, $q(x)$, $p(x)$ are continuous functions of x

λ is a constant.

• A non-zero solution of above BVP is called Eigen function. The value of λ in S.L. Eqn of above BVP, is called eigen value, if non-zero solution of BVP exists.

Example :-

$$y'' + \lambda y = 0, \quad \} \text{ find eigen values n. times.}$$

Example :-

SL equation
 $r(x) = 1$

$$p(x) = 1$$

$$q(x) = 0$$

$$\lambda = \lambda$$

$$\left\{ \begin{array}{l} y'' + \lambda y = 0 \\ y(0) = 0 \\ y(\pi) = 0 \end{array} \right.$$

find eigen values
and eigen functions.

$$y'' + \lambda y = 0 \quad ; \quad (D^2 + \lambda)y = 0$$

$$r^2 + \lambda = 0 \quad \Rightarrow \quad r^2 = -\lambda.$$

Ch Eqn

Case I :- $\lambda = -v^2 \quad ; \quad r^2 = -(-v^2) \Rightarrow r = \pm v \quad ; \quad \underline{v \in \mathbb{R}}.$

Complete solution

$$y = C_1 e^{vx} + C_2 e^{-vx}$$

$$y(0) = 0 \Rightarrow 0 = C_1 + C_2$$

$$y(\pi) = 0 \Rightarrow 0 = C_1 e^{\pi v} + C_2 e^{-\pi v} \quad \left. \begin{array}{l} \rightarrow C_1 = 0 \quad C_2 = 0 \\ \Rightarrow y = 0 \end{array} \right\}$$

$\Rightarrow \lambda = -v^2$ are Not eigen values.

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Case II: $\lambda = 0$, $r^2 = 0$, $r = 0, 0$ $y = C_1 e^{0x} + C_2 x e^{0x}$

Complete solution $y = C_1 + C_2 x$

$y(0) = 0 \Rightarrow C_1 = 0$

$y(\pi) = 0 \Rightarrow C_1 + C_2 \pi = 0 \Rightarrow C_2 = 0$
 $\Rightarrow y = 0$

$\Rightarrow \lambda = 0$ is Not an eigen value.

Case III: $\lambda = v^2$ ($v \in \mathbb{Z}$) $r^2 = -v^2 \Rightarrow r = \pm vi$

$y = C_1 \cos(vx) + C_2 \sin(vx)$

$y(0) = 0 \Rightarrow C_1 = 0$

$y(\pi) = 0 \Rightarrow \pm C_1 = 0$

$\therefore v \in \mathbb{Z}$

$y = C_2 \sin(vx)$ is a solution of BVP.

$\Rightarrow \lambda = v^2$, $v \in \mathbb{Z}$ is an eigen value, and the function is $\sin(vx)$.

$\Rightarrow \lambda = v^2, v \in \mathbb{Z} \Rightarrow \dots$
 corresponding eigen function is $\sin(vx)$.

• $\lambda = 1, 4, 9, 16, \dots$
 $\swarrow \quad \downarrow \quad \searrow \quad \rightarrow$
 $\sin x \quad \sin 2x \quad \sin 3x \quad \sin 4x$

• $\{ \sin(nx) \mid n \in \mathbb{Z} \}$ is set of eigen function of BVP
 w.o.t eigen values $\{ n^2 \mid n \in \mathbb{Z} \}$.

* Existence of Real eigen values.

we consider

$$(p(x)y')' + [q(x) + \lambda p(x)]y = 0 \quad ; \quad a \leq x \leq b$$

$$R_1 y(a) + R_2 y'(a) = 0$$

$$L_1 y(b) + L_2 y'(b) = 0$$

... .. real valued cont.

• If $p(x), q(x), r(x), r'(x)$, all are real valued cont. functions in $a \leq x \leq b$.

• $p(x) > 0$ for all $a \leq x \leq b$ of S.L. problem are real. Then all eigen values and Eigen functions of SL problem corresponding to distinct eigen values are orthogonal with respect to weight function $p(x)$.

* A set of functions $\{y_1(x), y_2(x), \dots\}$ defined on $a \leq x \leq b$ is called orthogonal w.r.t weight function $p(x)$ ($p(x) > 0$) if

$$\langle y_m(x), y_n(x) \rangle = \int_a^b p(x) y_n(x) y_m(x) dx = 0 \quad \text{if } m \neq n.$$

$$\langle y_m(x), y_n(x) \rangle = \int_a^b p(x) y_n(x) y_m(x) dx = 0 \quad \text{if } m \neq n.$$

• Check that the set $\{\sin(nx) \mid n \in \mathbb{Z}\}$ is an orthogonal set w.r.t weight function $p(x) = 1$ on interval $[0, \pi]$

$$\begin{aligned} \langle \sin(mx), \sin(nx) \rangle &= \int_0^\pi \sin(mx) \sin(nx) dx, \quad m \neq n \\ &= \frac{1}{2} \int_0^\pi [\cos(m-n)x - \cos(m+n)x] dx = 0 \end{aligned}$$

* Norm $\|y_n(x)\| = \sqrt{\langle y_n(x), y_n(x) \rangle} = \left(\int_a^b p(x) (y_n(x))^2 dx \right)^{1/2}$

• Norm of $\sin(nx)$ w.r.t weight function 1 and interval $[0, \pi]$
 $\frac{1}{\pi}$

$$\| \sin(nx) \| = \left(\int_0^\pi \sin^2(nx) dx \right)^{1/2} = \sqrt{\pi/2}$$

• Periodic Sturm Liouville's Problem

$$(r(x)y')' + (q(x) + \lambda p(x))y = 0$$

$$; \quad a \leq x \leq b$$

$$y(a) = y(b)$$

$$y'(a) = y'(b)$$

If $r(a) = r(b)$, then, eigen functions corresponding to distinct eigen values are orthogonal and all eigen values are real.

• Legendre's polynomial $\rightarrow$ solution of Legendre's Equation.

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0 \quad \text{--- ①} \quad -1 \leq x \leq 1$$

$$((1-x^2)y')' + n(n+1)y = 0$$

$$r(x) = 1-x^2, \quad p(x) = 1, \quad q(x) = 0, \quad \lambda = n(n+1)$$

- $P_n(x)$ is solution of ①.
- $P_n(x)$ are eigen functions corresponding to eigen value $n(n+1)$

eigen function	$\lambda = n(n+1)$
$P_0(x)$	0
$P_1(x)$	2
$P_2(x)$	6
$P_3(x)$	12

* $\{ P_n(x) \mid n \in \mathbb{Z}, n \geq 0 \}$ is orthogonal

set w. ot weight
function $p(x) = 1$.

n . < 1 Problem

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Proof of thm on Periodic S.L. Problem

(i) Eigen functions corresponding to distinct eigen values are orthogonal.

Let $y_m(x)$ and $y_n(x)$ be eigen function corresponding to eigen values λ_m and λ_n sep.
 $\lambda_m \neq \lambda_n$

T.P.

$$\int_a^b p(x) y_m(x) y_n(x) dx = 0$$

*

$$\left[r(x) y_m'(x) \right]' + \left[q(x) + \lambda_m p(x) \right] y_m(x) = 0$$

$$\left(r(x) y_m'(x) \right)' + q(x) y_m(x) = -\lambda_m p(x) y_m(x)$$

Similarly

$$(r(x) y_m'(x))' + q(x) y_m(x) = -\lambda_m p(x) y_m(x)$$

$$(r(x) y_n'(x))' + q(x) y_n(x) = -\lambda_n p(x) y_n(x)$$

$$\lambda_m - \lambda_n \int_a^b p(x) y_m(x) y_n(x) dx$$

$$= \int_a^b \underbrace{\lambda_m p(x) y_m(x)}_{\lambda_m p(x) y_m(x)} y_n(x) dx - \int_a^b \lambda_n p(x) y_m(x) y_n(x) dx$$

$$= \int_a^b \underbrace{(r(x) y_m'(x))' y_n(x) + q(x) y_m(x) y_n(x)}_{(r(x) y_m'(x))' y_n(x) + q(x) y_m(x) y_n(x)} dx$$

$$- \int_a^b \underbrace{(r(x) y_n'(x))' y_m(x) + q(x) y_n(x) y_m(x)}_{(r(x) y_n'(x))' y_m(x) + q(x) y_n(x) y_m(x)} dx$$

$$= r(x) y_m'(x) y_n(x) - r(x) y_n'(x) y_m(x) \Big|_a^b$$

$$r(a) = r(b)$$

$$= \lambda(b) \left[\cancel{y_m'(b) y_n(b)} - \cancel{y_n'(b) y_m(b)} - \cancel{y_m'(a) y_n(a)} + \cancel{y_n'(a) y_m(a)} \right]$$

$$\underline{\underline{\lambda(a) = \lambda(b)}}$$

$$= 0.$$

$$y_n(a) = y_n(b)$$

$$y_n'(b) = y_n'(a)$$

$$y_m(a) = y_m(b)$$

$$y_m'(a) = y_m'(b)$$

II :- Eigen values of periodic SL problem are real

Let $\lambda = \alpha + i\beta$ be an eigen value, and $y(x)$ be function corresponding eigen value λ .
 If possible

• eigen function corresponding to $\bar{\lambda}$ will be $\bar{y}(x)$.
 If λ is a complex number then $\lambda \neq \bar{\lambda}$.

If λ is a complex number then $\lambda \neq \bar{\lambda}$.
 (By first part)

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$$\lambda - \bar{\lambda} \int_a^b p(x) y(x) \bar{y}(x) dx = 0$$

(By first part
of this
thm)

$$\lambda - \bar{\lambda} \underbrace{\int_a^b p(x) |y(x)|^2 dx}_{\neq 0} = 0$$

$$\forall a \leq x \leq b$$

$$p(x) > 0$$

 $\Rightarrow$

$$\lambda - \bar{\lambda} = 0$$

$$\Rightarrow \lambda = \bar{\lambda}$$

$\Rightarrow \lambda$ is a real no.